

Deepansh Sabharwal

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Profile

Third-year Computer Engineering student who has done the full train, evaluate, deploy loop for object detection on edge hardware and then debugged it on the mounted hardware. Fine-tuned YOLOv8n, quantized it to INT8, deployed it on a Raspberry Pi with an NPU, and fixed the failures that only showed up on the mounted unit. Wants to be on the water when the test fails, not just in the simulator. Canadian permanent resident.

Available for 4, 8 or 12 months from January 2027. Would relocate to the Sunshine Coast.

Experience

University of British Columbia | Undergraduate Research Assistant **May 2026 - Present**
Edge computer vision, now signal processing / Python, OpenCV, Raspberry Pi 5, Hailo-8L NPU, Linux, Git

- Fine-tuned YOLOv8n for a wildlife collision warning system (mAP50 0.823, mAP50-95 0.638) and deployed it in real time on a Raspberry Pi 5 with a Hailo-8L NPU as a systemd service, verified by cold boot to detect within about 45 seconds of power-on
- Quantized the model to INT8 with a 2,001-image calibration set and quantization-aware fine-tuning; root-caused a compiler partition failure to the tool's automatic postprocess extraction and rebuilt the export path with explicit end-node parsing and a custom NMS config
- Integrated the camera on the mounted hardware and ran checkerboard calibration in OpenCV (1.34 px reprojection error); made the I2C display driver resilient to a flaky connector
- Fixed failures that only appeared on the physical unit: a signal handler overridden by a vendored library, a thread teardown race confirmed with a 20-trial stress test; maintain 47 unit tests behind a simulation backend

Janvani, Citizen Journalism Platform | Founder and Sole Developer **2026 - Present**
TypeScript, Node.js, PostgreSQL, Git, CI. Live at janvani-sable.vercel.app; every change shipped through automated CI

Best Buy Canada | Sales Advisor **Jul 2025 - Present**
Kelowna, BC. Customer-facing technical role, 20+ consultations daily, explaining technical options to non-technical customers

Projects

AI-in-Farming: Crop Disease Classifier | Solo **2025**
Python, TensorFlow, NumPy, pandas

- Trained a convolutional classifier on about 15,000 leaf images to 92% test accuracy, improving it through confusion analysis rather than blind hyperparameter search

ThermoGuard Safety System | UBC team of 4 **2026**
C, Arduino, Raspberry Pi. Real-time control firmware over UART, I2C, SPI; isolated bring-up faults at the hardware/software boundary

Technical Skills

Languages: Python (strong), C (strong), Bash, TypeScript, Verilog HDL

Perception: YOLO-family fine-tuning and evaluation (mAP), dataset building and calibration sets, INT8 quantization for an NPU, OpenCV camera calibration, real-time pipelines on embedded Linux

Hardware and field: Camera and sensor integration over I2C, SPI, UART, GPIO; systemd deployment; cold-boot verification; stress testing; root cause from logs on the physical unit

Tooling: Git, GitHub Actions CI, gdb, Linux, Claude Code and Copilot as daily coding agents

Education

University of British Columbia, Okanagan | B.A.Sc. Computer Engineering, expected May 2029 Kelowna, BC
Coursework: Microcomputer Engineering, Microelectronics I, Linear Circuit Theory, Industrial Automation, Data Structures and Algorithms (Java)